

Virtual Characterization Lab Toolkit

Preprocessing

Since the current version of LAMMPS supports either charge or spin but not both simultaneously, if your file initially has charge or spin and you aim to add spin or charge, the charge/spin will be eliminated and then spin or charge will be assigned. The default assigned charge and spin values are zero for your convenience, you can add charge and spin to the atoms within the LAMMPS script itself using the `set` command.

LAMMPS scripts

Several LAMMPS scripts are available for spin-lattice dynamics, ReaxFF, and other simulations. These can be modified and run based on your study's purposes.

⚠ Note: This repository of the lammps input files is intended primarily as a collection of examples. For accurate materials simulations, users should refine and adjust the parameters (e.g., magnetic and mechanical interactions) to suit their specific systems.

SAED and XRD patterns cannot be calculated for triclinic cells using LAMMPS. One should first finalize the calculation, then convert the desired/last snapshot of the system (which is stored as a .cfg file) to the LAMMPS format using the VCL toolkit. You can then use the auxiliary LAMMPS script named `in.ax` to obtain the XRD and SAED patterns.

For spin-lattice dynamics, you need to adjust the system size and the number of cores used in the calculation. If your system size is too small or you use too many cores, you will receive the following error.

```
ERROR: Illegal sectoring operation (src/SPIN/fix_nve_spin.cpp:601)
```

To avoid this error, you should either reduce the number of cores or increase your system size. It is recommended to increase the system size. For more information, please see the main article.

```
article{tranchida2018massively,  
title={Massively Parallel Symplectic Algorithm for Coupled Magnetic Spin Dynamics and Molecular  
Dynamics},  
author={Tranchida, J and Plimpton, S J and Thibaudau, P and Thompson, A P},  
journal={Journal of Computational Physics},  
volume={372},  
pages={406--425},  
year={2018},  
publisher={Elsevier}  
doi={10.1016/j.jcp.2018.06.042}  
}
```

Selected area electron diffraction (SAED)

To have a 3D analysis of your system, you should set the zone axis of the incident radiation (the zone value) to zero, i.e., 'Zone 0 0 0'. This is specifically useful for analyzing defects. You can then use different slices along various orientations to fully analyze your desired region by changing the parameters below in VCL toolkit, which are set to provide a view along the X direction in the post-processing section for SAED pattern visualization.

spherical slice origin [39.84063, 0, 0]:
view normal vector (e.g., '-1, 0, 0') [-1, 0, 0]:
view up vector (e.g., '0, 1, 0') [0, 1, 0]:

for more information see 'Slice operator' in:

https://visit-sphinx-github-user-manual.readthedocs.io/en/v3.2.2/gui_manual/Operators/OperatorTypes/Slice_operator.html

However, setting Zone values to zero may require a huge amount of memory for calculations and can lead to a very large `.vtk` output file. The visualization of these large `.vtk` files also requires parallel processing and significant memory.

To avoid this, you can:

- Set the **Zone** value based only on your desired view.
- Use the `compute saed` and `saed/vtk` commands multiple times in a simulation along different directions.
- Specify only a small region for SAED analysis.